12-7-2026

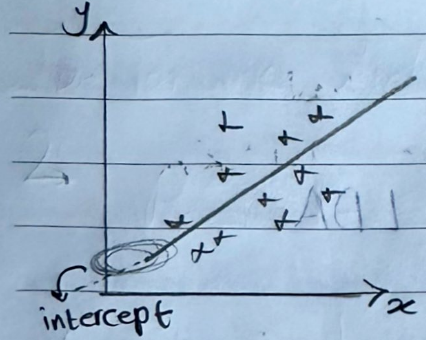
التاريخ: / /   الموضوع:

linear Regression

* linear Regression :

⇒ try to find the Best fit line which will helps us to do the prediction

formula : $h_\theta(x^i) = \theta_0 + \theta_1 x$ or $y = mx + c$ ...



θ_0 = intercept
θ_1 = slop or coefficent
x = data points

(θ_0, θ_1) ↳ also called weights

⇒ the Best fit line is when the distane beltween Predicted point "on the line" and data points is minimum.

* Cost function :

⇒ used to Show how well the model is perforing by measuring the difference between the predicted Value and the actual data

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} (\underbrace{h_\theta(x^i)}_{\theta_0 + \theta_1 x} - y^i)^2$$

↳ cost function algorithm (MSE . mean square error)

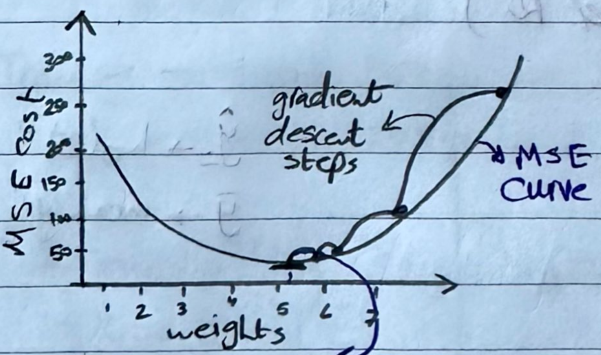
بنقسم على m عشان نجيب المتوسط لمربعات الأخطاء (m no of points)

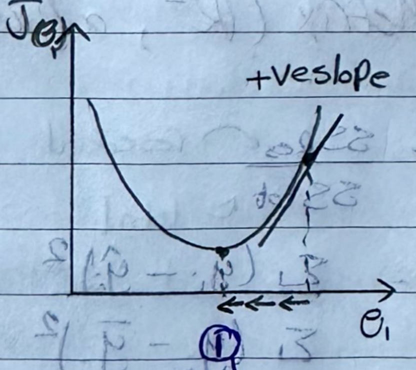
وضربنا في ½ عشان نسهل عملية الاشتقاق لما نعمل update للأوزان (θ_0, θ_1)

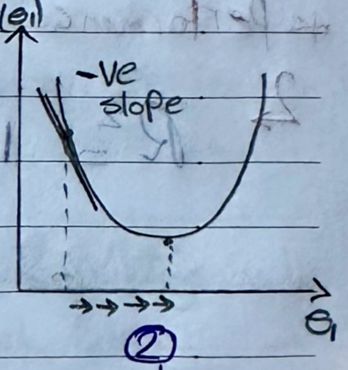
* Gradient Descent

↳ an optimization algorithm used to minimize the cost function, iteratively adjust the weights of the model to Reduce error

↳ each iteration updates the weights in the direction that minimize the cost function ⊕ leading to the optimal set of parameters







Global minimum هو دة الي المفروض اوصله

* Convergence algorithm "Repeat untill Convergence":

↳ $\{ \theta_j = \theta_j - \alpha \boxed{\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)} \}$ Partial diff.

learning rate

at ① → $\theta_{new} = \theta_{old} - \alpha(+ve\ slope)$ هينقص عشان يوصل لل global minimum

② → $\theta_{new} = \theta_{old} - \alpha(-ve\ slope)$ هيزود ,, ,, ,,

Gradient descent

→ "Repeat untill convergence"

{

$$\theta_0 = \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(h_\theta(x^i) - y^i\right)$$ بعد ما نخلص

$$\theta_1 = \theta_1 - \underset{\text{Learning rate}}{\alpha} \frac{1}{m} \sum_{i=1}^{m} \left(h_\theta(x^i) - y^{(i)}\right) x^i$$

}

* Performance Metrix (R^2, adjusted R^2)

→ $$R^2 = 1 - \frac{SS_{Res} \leftarrow \text{resedual}}{SS_{tot} \leftarrow \text{total}}$$

$\hat{y}_i \rightarrow h_\theta(x^i)$

$\bar{y} \rightarrow$ mean y

$$= 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y}_i)^2}$$

* R^2 disadvantages:-

لو أضفت features ملهاش علاقة بالهدف اللي عايزة أتوقعه بتحل عليها وتطلع accuracy جديد أعلى زي مثلا معايا عدد غرف شقة وعايزة أتوقع سعرها وإداني acc 90% وأضفت feature زي موقع الشقة إداني accuracy 92% بس لو أضفت مثلا Gender ودة ملهوش علاقة بتوقع سعر الشقة للأسف R^2 هيدي عليها acc. مثلا 93% ودة غلط

الحل في ← adjusted R^2

* R^2 adjusted:-

$$= \frac{1 - (1-R^2)(N-1)}{N-P-1}$$

* P = num-of features or predectors

→ and so it neglects any feature that is not correlated

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التاريخ: / /　　　　　　　　　　　　　　　　　　　　　　　　　　　　الموضوع:

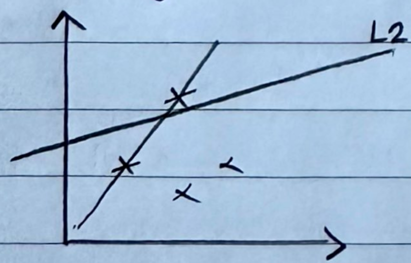
Ridge and lasso Regression

over fitting → the model performs well on traing data but fails test data

under fitting → the model gives bad accuracy with both train and test data

Model 1	Model 2	Model 3
Training accuracy = 90%	" = 90%	= 70%
test accuracy = 76%	= 91%	= 65%
⇒ overfitting	⇒ Generalized model	⇒ under fitting
↳ low Bias	Low Bias	High Bias
↳ High Variance	Low Variance	High Variance

* Ridge "L_2 regularization":



$$(\hat{y}-y)^2 + \lambda\,(\text{slop})^2$$

$\lambda \rightarrow$ is a hyper parameter to check how fast you want to lessen the steepness to grow higher

L_2: used to overcome overfitting problem, through modifiying the cost function by adding a penalty that lessen the steepnes of slope or reduces the weight and stop them from getting larger

Lasso Regression "L_1 Regularization"

↳ $(\hat{y}-y)^2 + \lambda |slope|$ → Absolute value ↓ helps in Feature selection

in L_2 → the Value will be reduced but never reachs Zero

in L_1 → the Value will be reduced and may be Zero.

L_1. cost func. $= (h_\theta(x^i) - y^i)^2 + \lambda |slope|$

⇒ prevents overfitting, and helps with Feature selection

L_2. cost func. $= (h_\theta(x^i) - y^i)^2 + \lambda (slop)^2$

⇒ prevents overfitting